

The Wave-Speed Hinge

One kinematic relation, three different pieces of medium physics

Working rule: For a monochromatic travelling wave, $v_p = f\lambda$ relates phase speed, frequency and wavelength. It does not determine the speed. The source and boundary conditions select a frequency; the medium law supplies v_p ; the wavelength follows.

Name: _____

Class: _____

Companion task to *The Placeholder Force*

1. A transverse wave on a stretched string

A driver oscillates one end of a long string at $f = 80\text{ Hz}$. The string is under tension $T = 72\text{ N}$ and has linear mass density $\mu = 2.0 \times 10^{-3}\text{ kg m}^{-1}$. Ignore damping and dispersion.

A. Represent the travelling disturbance

Sketch several crests and troughs. Mark the direction of propagation, one wavelength, and the transverse motion of a small element of string. Distinguish particle motion from wave motion.

B. Supply the medium law

Use

$$v = \sqrt{\frac{T}{\mu}}$$

to calculate the wave speed. Identify the restoring property and the inertial property. Explain why neither the driver frequency nor $v = f\lambda$ determines this value.

C. Apply the kinematic hinge

Use $v = f\lambda$ to calculate the wavelength. The driver frequency is then doubled without changing the string. Predict the new v and λ , and justify each answer.

D. Interrogate the closure

Predict the effect on wave speed of quadrupling T and, separately, quadrupling μ . Which equation contains this information? What idealisations fail for a real string at very high frequency or large amplitude?

2. A longitudinal sound wave in water

An ultrasonic transducer emits a 40 kHz sound wave into water. Take the water density as $\rho = 1.00 \times 10^3\text{ kg m}^{-3}$ and its bulk modulus as $B_{\text{bulk}} = 2.20 \times 10^9\text{ Pa}$.

A. Represent compression and propagation

Sketch compressions and rarefactions. Mark one wavelength, the propagation direction, and the back-and-forth displacement of fluid elements. Explain why the wave is longitudinal even though energy moves forward.

B. Close the model with fluid properties

Use

$$v = \sqrt{\frac{B_{\text{bulk}}}{\rho}}$$

to calculate the sound speed. Interpret B_{bulk} as resistance to compression and ρ as inertia. Why does a larger bulk modulus increase v , while a larger density decreases it?

C. Join speed, frequency and wavelength

Calculate the wavelength. The same transducer now sends the same frequency into a fluid with a lower wave speed. State what remains continuous at the boundary and what must change. Explain without saying that the wave “slows because its frequency falls”.

D. Interrogate the model

The formula treats the fluid as uniform, continuous and linearly elastic. Which features of a gas or liquid are compressed into B_{bulk} and ρ ? What additional effect would make speed depend on frequency?

3. Light entering glass

A monochromatic laser has vacuum wavelength $\lambda_0 = 600\text{ nm}$. It enters ordinary glass normally. At this frequency the refractive index is $n = 1.50$. Take $c = 3.00 \times 10^8\text{ m s}^{-1}$.

A. Represent the boundary

Draw wavefronts before and inside the glass. Mark the boundary and propagation direction. Because incidence is normal, the ray does not bend; nevertheless the spacing of the wavefronts changes. Show this change.

B. Supply the optical medium law

Use

$$v = \frac{c}{n}$$

to calculate the phase speed in glass. Explain what the refractive index packages about the electromagnetic response of the material. Why is n , rather than $v = f\lambda$, the empirical closure here?

C. Apply the hinge across the boundary

Calculate the laser frequency in vacuum and its wavelength in glass. State explicitly why the frequency is unchanged at a stationary boundary while the wavelength changes. Verify that $\lambda = \lambda_0/n$.

D. Which wave speed?

In a dispersive material, $n = n(f)$. Explain why different frequencies can then have different phase speeds. Distinguish, qualitatively, phase velocity from the group or signal velocity of a pulse.

Synthesis: $v = f\lambda$ organises the wave; the medium law determines its speed

1. Complete the comparison.

Wave	Restoring property	Inertial property	Speed law	What fixes f ?
String				
Sound in water				
Light in glass				

2. Develop the central claim.

Respond to:

“The equation $v = f\lambda$ is kinematic bookkeeping. It coordinates a wave’s pattern in space and time but does not explain why that pattern travels at that speed.”

Use all three cases. Include the terms *source*, *medium*, *restoring response*, *inertia*, *boundary condition*, *phase velocity*, *wavelength*, and *empirical closure*. Explain what changes, and what remains fixed, when a wave crosses into a new medium.

Teacher guide and indicative answers

The Wave-Speed Hinge

1. String: restoring tension and distributed inertia

Representation. Elements of string move transversely while the pattern and energy propagate along the string. Wavelength is the spatial period of the pattern.

Medium speed.

$$v = \sqrt{\frac{72}{2.0 \times 10^{-3}}} = 1.90 \times 10^2 \text{ m s}^{-1}.$$

Tension supplies the restoring action; linear density supplies inertia. The speed law comes from the string dynamics, not from the driver or the identity $v = f\lambda$.

Wavelength.

$$\lambda = \frac{v}{f} = \frac{189.7}{80} = 2.37 \text{ m}.$$

If f doubles, the same ideal nondispersive string still gives the same v , so λ halves to 1.19 m.

Parameter changes. Since $v \propto \sqrt{T}$, quadrupling tension doubles v . Since $v \propto 1/\sqrt{\mu}$, quadrupling linear density halves v . At large amplitude, tension may vary during the motion; at high frequencies, bending stiffness, damping or non-uniformity can introduce dispersion.

Conceptual emphasis. $v = f\lambda$ can calculate any one quantity once the other two are known. It contains no information about why tension and mass density select the speed.

2. Sound: compressibility and mass density

Representation. Fluid elements oscillate parallel to the propagation direction. Compressions and rarefactions move; individual elements do not travel with the wave over large distances.

Medium speed.

$$v = \sqrt{\frac{2.20 \times 10^9}{1.00 \times 10^3}} = 1.48 \times 10^3 \text{ m s}^{-1}.$$

The bulk modulus is the restoring stiffness against compression. Density measures the inertia that must be accelerated. A stiffer medium restores disturbances more rapidly; a denser medium responds more slowly for the same restoring stress.

Wavelength.

$$\lambda = \frac{1483}{4.0 \times 10^4} = 3.71 \times 10^{-2} \text{ m}.$$

At a stationary boundary, the temporal oscillation must match on both sides, so the frequency remains that of the source. A lower speed therefore produces a shorter wavelength.

Model limits. B_{bulk} packages the pressure-density response and depends on thermodynamic conditions. Temperature, composition, molecular relaxation and other frequency-dependent responses can produce dispersion and attenuation.

Conceptual emphasis. The continuum law closes the model. The kinematic hinge then translates the selected speed into a wavelength for the imposed frequency.

3. Light: material response expressed by refractive index

Optical speed.

$$v = \frac{3.00 \times 10^8}{1.50} = 2.00 \times 10^8 \text{ m s}^{-1}.$$

The refractive index summarises the material's electromagnetic response at the stated frequency. More fundamentally, the phase speed follows from the relevant electric and magnetic constitutive response.

Frequency and wavelength.

$$f = \frac{c}{\lambda_0} = \frac{3.00 \times 10^8}{600 \times 10^{-9}} = 5.00 \times 10^{14} \text{ Hz},$$

$$\lambda = \frac{v}{f} = 4.00 \times 10^{-7} \text{ m} = 400 \text{ nm} = \frac{\lambda_0}{n}.$$

The field oscillations must remain phase-matched in time at a stationary interface; therefore frequency is continuous. Speed changes because the medium changes, so the spatial period changes.

Dispersion. If $n = n(f)$, each monochromatic component has $v_{\text{p}} = c/n(f) = f\lambda$. A pulse is a superposition of components. The envelope or information typically travels at the group velocity $v_{\text{g}} = d\omega/dk$, which need not equal the phase velocity.

Conceptual emphasis. $v = f\lambda$ remains true for each monochromatic component, but it does not supply $n(f)$ or distinguish phase speed from pulse speed.

Teacher synthesis: expected conceptual structure

Wave	Restoring property	Inertial property	Speed law	Source/boundary role
String	tension T	linear density μ	$\sqrt{T/\mu}$	driver fixes f
Sound	bulk modulus B_{bulk}	density ρ	$\sqrt{B_{\text{bulk}}/\rho}$	transducer fixes f
Light	electromagnetic material response, encoded by $n(f)$	field-material response is already packaged into n	c/n	stationary boundary preserves f

A strong response should state that $v = f\lambda$ is a compatibility relation between a wave's temporal period and spatial period. It is not a constitutive or dynamical law. The string, acoustic and optical speed laws encode distinct physical mechanisms.

When a monochromatic wave crosses a stationary boundary, the source frequency remains continuous. The new medium selects a new phase speed, and the wavelength adjusts. In a dispersive medium, the index or material parameters depend on frequency, and phase and group velocities must be distinguished.

Diagnostic misconception: “The frequency decreases because the wave slows.” Correction: for a fixed source and stationary boundary, the frequency remains fixed; the wavelength changes.